

my boss app — Android Studio & Mobile App Run Guide

Audience: Developers, QA

Purpose: Run the Flutter employee app and connect it to the backend

1. Prerequisites

Tool	Version
Android Studio	Latest stable
Flutter SDK	3.35.7 / Dart 3.9.2 (FVM — see <code>apps/mobile/.fvmrc</code>)
Backend	Running (see <code>02_RUN_DEMO_SERVER.md</code>)

Verify Flutter:

```
fvm flutter doctor
```

2. Open project in Android Studio

1. Launch **Android Studio**

1. **File → Open**

1. Select folder: `my_boss_v5/apps/mobile`

1. Wait for Gradle sync to finish

1. If prompted: **Trust Project**

3. Start Android Emulator

1. Tools → Device Manager

1. Create device (if none): Pixel 6, API 34 recommended
 1. Click ► **Play** to start emulator
 1. In Android Studio toolbar, select the emulator as run target
-

4. Start backend (required before mobile)

Option A — Docker (demo / infra):

```
cd my_boss_v5
./infrastructure/scripts/deploy-demo-server.sh 127.0.0.1
./infrastructure/scripts/verify-mobile-api.sh 127.0.0.1 --gateway
```

Option B — Local dev (no Docker):

```
cd my_boss_v5/apps/backend
npm run start:dev
```

5. Run mobile app

Terminal (or Android Studio **Run** ► button):

```
cd my_boss_v5/apps/mobile
fvm flutter pub get
fvm flutter gen-l10n
fvm flutter run --dart-define=DEMO_MODE=true
```

Or in Android Studio: select `main.dart` → click green **Run** button (add `--dart-define=DEMO_MODE=true` in Run Configuration).

6. API host modes

Mode	Command	Backend path
Demo (recommended)	<code>--dart-define=DEMO_MODE=true</code>	Gateway :8090 (probed at startup)
Dev emulator	<code>--dart-define=ENV=development</code>	Direct ports 3001–3005 via 10.0.2.2
Dev physical	<code>--dart-define=API_HOST=<lan-ip></code>	Direct ports 3001–3005

When `DEMO_MODE=true`, the app probes `GATEWAY_ORIGIN` and `API_HOSTS` and picks the first reachable `/health`.

Emulator with demo gateway:

```
fvm flutter run --dart-define=DEMO_MODE=true --dart-define=GATEWAY_ORIGIN=http://10.0.2.2:8090
```

7. Physical Android device

Demo APK (recommended for QA / external devices)

```
cd apps/mobile
./build-local-android.sh
# Output: build/android-dist/myboss-demo-<lan-ip>.apk
```

Copy APK to phone, uninstall old app first, then install.

Requirements:

- Phone and Mac on same Wi-Fi **or** Cloudflare tunnel running
- Demo gateway on port **8090**

Development run (direct microservice ports)

1. Find your computer IP:

```
ipconfig getifaddr en0 # macOS example: 10.6.208.129
```

1. Run:

```
fvm flutter run --dart-define=ENV=development --dart-define=API_HOST=<lan-ip>
```

1. Phone and laptop must be on the **same Wi-Fi**. Allow firewall access to ports **3001–3005** (dev) or **8090** (demo).

8. Demo login

Field	Value
Email	demo@orange.com
OTP	Auto-filled when DEMO_MODE=true

Other test emails: nisreen.a@orange.com , omar.t@orange.com

9. Release demo APK

```
cd apps/mobile
./build-local-android.sh
# or
./build-demo-apk.sh
```

Output: build/android-dist/myboss-demo-<lan-ip>.apk or
build/app/outputs/flutter-apk/app-release.apk

10. Common issues

Problem	Fix
Network error on login	Backend not running; check <code>curl http://127.0.0.1:8090/health</code>
Emulator cannot reach API	Use <code>GATEWAY_ORIGIN=http://10.0.2.2:8090</code> with <code>DEMO_MODE=true</code>
Physical device fails	Same Wi-Fi; test <code>http://<mac-ip>:8090/health</code> in phone browser
Old APK still fails	Uninstall app before installing new APK
Gradle sync failed	File → Invalidate Caches; <code>fvm flutter clean && fvm flutter pub get</code>
OTP not received	Check auth logs for <code>[DEMO OTP]</code> ; use demo build with auto-fill

Orange — my boss app — Mobile

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